Transfusion and Apheresis Science 63 (2024) 103935

Contents lists available at ScienceDirect

Transfusion and Apheresis Science

journal homepage: www.elsevier.com/locate/transci

Focus on: Prognostic scores to predict stem cell mobilization

Attilio Olivieri *, Francesco Saraceni, Amalia De Luca [1]

Dept &Stem Cell Transplant Unit, DISCLIMO-Università Politecnica delle Marche, Ospedali Riuniti di Ancona, Ancona, Italy

Autologous stem cell transplantation (ASCT) still represents a mainstay of treatment for multiple myeloma (MM) patients eligible for ASCT, late relapses of Non Hodgkin Lymphoma (NHL) and Hodgkin disease (HD). ASCT is almost exclusively performed today with peripheral blood stem cells (PBSCs) mobilization; therefore, stem cell mobilization currently represents a crucial step; despite developments in the mobilization procedures, a proportion of patients between 5% and 30% fail to collect an adequate number of CD34 + cells. Poor mobilization forces the patient to undergo a re-mobilization procedure and, in some cases, leads to postponing or even abandoning a transplant strategy. Several factors have been associated with poor mobilization [1]; however, an exhaustive profile of the patient at risk of sub-optimal mobilization is still missing.

Hematopoietic stem cells (HSC) may be mobilized in peripheral blood (PB) by chemotherapy alone, haematopoietic growth factors alone or by chemotherapy followed by haematopoietic growth factors. G-CSF alone is still the golden standard for PBSC mobilization in healthy donors while in lymphoma or MM patients G-CSF can be administered alone or after mobilizing chemotherapy [2].

G-CSF is a naturally occurring cytokine, whose levels are significantly increased in healthy individuals in response to physiologic stimuli, primarily infections. After administration of a single 300-mcg dose of filgrastim to healthy volunteers, peak G-CSF plasma levels were observed at 4 h after injection (25,000 pg/mL−44,000 pg/mL), meaning a 500- to 1000-fold increase and returned to pre-injection values after 48 h. Commonly reported adverse effects of G-CSF are generally mild; severe side effects requiring G-CSF discontinuation are uncommon (1% to 3% of donors) [3–6]. G-CSF may induce a transient prothrombotic or hypercoagulable state in some healthy donors [7,8] and rare donors with vascular events have been reported during PBSC mobilization.

G-CSF does not directly mobilize HSC and progenitors by a direct influence on these cells. G-CSF blocks the interaction of CXCL12 and CXCR4 and increases the expression of independent transcriptional repression growth factor 1 (Gfi-1) so that it cuts down the level of CXCR4. G-CSF likely mediates cell mobilization through many other mechanisms, including disruption of VCAM-1/VLA-4interaction, cleavage of CXCR4 and CD44 expressed on HSCs and HPCs and cleavage of SDF-1 on some niche cells, as well as enhancement of the sympathetic nervous system mediated-mobilization [9].

The Ideal PBSC collection should be characterized by: 1)large amount of collected CD34 + (enough for two ASCT procedures at least in high risk MM patients);2)to be accomplished in one short leucapheresis (LK) procedure; 3)without needing several days of monitoring; 4) without reaching exaggerated leukocyte count; 5)low neutrophil contamination;6)high immunocompetent cell content [10]; 7)easy to plan LK (fixed collection day;8)no need of toxic mobilizing agents, avoiding severe adverse events during mobilization/collection.

The minimal threshold CD34 + cell dose to be infused is agreed to be ≥ 2–2.5 million CD34 cells/kg for a single ASCT; higher doses (5million CD34 cells/kg) are associated with long term stable engraftment, fast platelet and neutrophil engraftment, reduction in the need for supportive measures, leading to a significant cost sparing reduced toxicity and increased survival rates [11].

Among the possible defects causing poor **mobilization** (Table 1) we can distinguish between technical (1) and biological factors (2) such as: 1-inadequate dose of G-CSF or timing of collections; inadequate venous access; insufficient blood volume processed; presence of paraproteins etc); 2) insufficient number of HSC; damaged/inadequate microenvironment: defective niches; low response of effector/supporter cells such as macrophages or β-adrenergic nerves. These possible defects are not mutually exclusive. Besides these general factors other specific factors can affect SC mobilization. Malignant Bone Marrow involvement is often associated with poor mobilization, due to impairment of healthy niches by malignant cells in the BM or direct competition between HSC and malignant cells. Indolent lymphoproliferative diseases, and acute myeloid leukemia have also been identified as independent risk factors [12–15].

Radiotherapy, cytotoxic treatments and some targeted therapies can have deleterious effects on HSC and the niches and have been associated with an increased risk of mobilization failure.

Prior radiotherapy to large areas of red marrow can be associated with mobilization failure due to several combined effects: direct HSC toxicity and toxicity to the niche, damaging sinusoidal vessels and

* Corresponding author.

E-mail addresses: Attilio.Olivieri@ospedaliriuniti.marche.it (A. Olivieri), francesco.saraceni@ospedaliriuniti.marche.it (F. Saraceni), amalia.deluca@ospedaliriuniti.marche.it (A. De Luca).

[1] Co-author.

https://doi.org/10.1016/j.transci.2024.103935

Available online 23 April 2024

periarteriolar niches [16,17] and causing a "fatty" marrow, like the changes associated with aging [18,19]. Radiotherapy may also increase the expression of protease inhibitors such as α1-antitrypsin that would diminish the protease storm during mobilization [20,21].

Like radiation exposure, chemotherapy can cause HSC depletion and niche damage, including injury to mesencymal stem cells (MSC), osteoprogenitors, and the sympathetic nervous system [22,23]. DNA cross-link agents such as melphalan or carmustine and purine analogs such as fludarabine are associated with high risk of mobilization failure [24,25]. Several lines of chemotherapy may also damage niches for HSC and BM macrophages [26]. Lenalidomide may suppress HSC motility similar to the way it reduces the motility of marrow endothelial cells in multiple myeloma; the antiangiogenic effect of lenalidomide could also impair mobilization [27,28]. Recently, Daratumumab administration in MM patients has also been associated with a significant decrease in the number of collected HPC [29].

Poor mobilization is more frequently observed in elderly patients, but also in healthy donors over 50 years, probably due to a progressive telomere shortening and reduction in HSC number; in this setting endosteal osteoblastic niches for HSC can be damaged due to a progressive osteoblast reduction.

In spite of several retrospective studies suggesting a higher rate of mobilization failure (or suboptimal HSC yeld) among older patients, both in the lymphoma and in MM setting, the impact of age remains uncertain as well as the identification of a clear age cut-off [30–32].

The observation of a suboptimal mobilization among some healthy donors suggests the influence of genetic factors. Some genetic polymorphisms such as SDF-1, CD44, have been associated with suboptimal mobilization; moreover, the mannan-binding lectin deficiency, a form of complement deficiency, observed in about 10% of the population, could be also responsible for poor mobilization [33–36].

Diabetes mellitus is another risk factor for poor cell mobilization; the bone marrow niches of diabetic patients have altered microvasculature and perivascular sympathetic nervous systems with impaired MSC function and decreased osteoblasts [37].

Clonal haematopoiesis of indeterminant potential (CHIP) increases in frequency with age; a recent study identified the presence of somatic mutations in ASXL1, DNMT3A, JAK2, SF3B1, TET2 and TP53 in CD34 + haematopoietic progenitor cell-apheresis products of 96 patients who undergo HSC mobilization for ASCT, with a significantly greater proportion of patients experiencing poor mobilization [38]. More recently, in 90 patients evaluated during HSC collection for ASCT, CHIP at low VAF $\geq$ 1% was common at the time of mobilization, in poor mobilizers as well as in matched controls. However, the distribution of specific mutations was different between both groups: TP53 mutations and PPM1D mutations were mainly detected in patients with poor mobilization potential and the incidence of t-MN was significantly higher in poor mobilizers [39].

The most commonly accepted criteria to define a successful CD34 + cell mobilization are the peak of CD34 + cells in PB and/or cumulative aphaeresis yield (in a single attempt or with a pre-fixed number of aphaeresis days); surrogate end points such as the percent candidate patients undergoing ASCT and transplant outcome engraftment kinetics, have also been proposed [40],[12].

The Gruppo Italiano Trapianto Midollo Osseo (GITMO) proposed a definition of the proven poor mobilizer (PPM) and the predicted poor mobilizer (pPM), adopting a consensus based on an analytic hierarchy process[11]. Conceptually, the issue of "poor mobilization" may pertain to three sequential phases: 1-first phase (predicted PM) including baseline parameters and the pivotal clinical data; 2-dynamic phase, when the mobilization procedure is ongoing, reflecting the biologic capability of the patient to mobilize SC (potentially subjected to changes: e.g. increasing the dose of G-CSF or addition of Plerixafor:P); 3-the final phase involves the interaction between the patient's SC mobilization ability and the performance of the apheresis procedure, including the timing, the apheresis volumes and the collection efficiency.

Currently, a low circulating CD34 + count before apheresis is widely accepted as the stronger parameter able to predict mobilization failure: thus, to assist the clinician in a timely and cost-effective use of P, various algorithms were developed, mainly based on the circulating CD34 + at day 4 (in case of steady-state mobilization with G-CSF alone) or at the time of leukocytes recovery (in case of chemo-mobilization)[15]. However, such algorithms are applicable only a few hours before the start of the apheresis procedure, while ideally the identification of patients at high risk of inadequate mobilization should be performed before starting the mobilization process. Such a tailored approach might help to choose the best mobilization strategy, optimizing resources management, avoiding the need for re-mobilization, and redundant days of apheresis. Remobilization attempts presented, in the past, a quite popular procedure, due to the lack of effective dynamic rescue strategies. Re–mobilisation using conventional approaches before the "Plerixafor era" failed to mobilise sufficient CD34 + cells for ASCT in about 30% of patients, even when the HSC obtained at re–mobilisation were pooled with previously cryopreserved PBSC from first mobilisation [41].

Remobilization success rates in the Plerixafor era are not clear, due to variable end points established in the prospective studies; however, both the AMD3100–3102 study in MM and the AMD3100–3101 study in NHL, showed a clinically meaningful benefit for those patients randomized to be rescued with G-CSF+P, with 96% and 88% of patient mobilized who successfully undergo ASCT, respectively.

In order to avoid the negative impact of a remobilization attempt, alternative approaches with P have been proposed before the evidence of mobilization failure:

The pre–emptive use of Plerixafor may have advantages in terms of

Table 1
Main predictors of poor CD34 + mobilization in patients with MM and NHL and in healthy donors.

Failed previous mobilization attempt[11]
Previous extensive radiotherapy to marrow bearing tissue[11]
Several full courses of previous chemotherapies potentially affecting stem cell mobilization[11]
Previous administration of Daratumumab[29,32]
Previous treatment with stem cell poisons (lenalidomide, Fludarabine, Melphalan)[11,27,28]
Advanced phase disease (>=2 previous cytotoxic lines)[11]
Refractory disease[11]
Extensive BM involvement at mobilization[16]
BM cellularity < 30% at mobilization[11,40]
Any cytopenia before mobilization (e.g. Plt count<70.000/mcl)[15]
Advanced age*[11,30,15]
Diabetes mellitus*[37]
Suboptimal dose of G-CSF (<10-12 mcg/day)*[12]
Poor collection at first leukapheresis (<1x10e6 CD34 +/kg)*[15]
Low CD34 + /WBC ratio the day before leukapheresis*[45]
Circulating CD 34 + in PB< 20/μl before starting leukapheresis[12]
Circulating CD34 + in PB< 5/μl before Plerixafor administration[15,43,46]

* predictor of poor mobilization in healthy donors

avoidance of cancelled apheresis and/or transplant slots, and also in terms of avoiding the negative quality–of–life impact of a failed mobilisation procedure. However, this approach for some pPM [42] may be affected by the risk of over-treatment with a negative costs-benefit impact.

P on demand could be more attractive as this precision approach could avoid an unuseful administration in patients who could successfully mobilize HSC without P; this approach could be established on 3 basic algorithms:

1-CD34-PB count kinetics, after G-CSF or G-CSF/chemo-based mobilization; 2-Implementing the CD34-PB count kinetics with some established risk factors; 3-The combination of CD34-PB+ count with WBC count (CD34 +/WBC ratio). Moreover the results of 1st LK harvest < 1x10e6 CD34 + /kg is a commonly accepted supplementary criterion for P addition. Unfortunately, such algorithms have been tested in the context of monocentric experiences or applied in the context of specific schedules of mobilization [43–46].

We conducted a retrospective analysis, including 1318 MM and lymphoma patients receiving mobilization procedures, to validate the predictive ability of GITMO criteria; we also elaborated a "poor mobilization risk score" (PMscore) easily applicable in real life, to help decision-making and procedure customization based on premobilization parameters. For each case, a score was generated by summing one point for each minor criteria and two points for each major criteria that were present. The new score (AUC=0.80) in the simplified version allowed us to indentify "a priori" those patients at very high risk of mobilization: indeed those with a PM score > 6.5 failed mobilization with a good positive predictive value (PPV = 71%).

Today, the first key decision in scheduling a first-line mobilization strategy, is the choice between a chemo-mobilization or a cytokine-only strategy. The second crucial stage is the identification of those patients, in whom the addition of just-in-time P could be useful and cost-effective. To this end, different algorithms have been proposed: all of them include a PB CD34 + cell count, the most reliable parameter to trigger plerixafor administration [47,48].

Nevertheless, such algorithms present several limits, leaving the unresolved a "gray zone" with intermediate values of PB CD34 + (i.e. 10–20/mcl), where no recommendations can be drawn and a "case by case" approach is suggested [49]. Recently published US recommendations [50], suggest to tailor the mobilization plan according to patient and disease characteristics; in the case of MM, the authors suggest chemo-mobilization for patients previously treated with lenalidomide. Similarly, for a patient with lymphoma, the authors recommend to limit steady-state mobilization (only G-CSF) to patients "at low risk for mobilization failure". However, the availability of new drugs or new approaches requires updating the profile of the so called pPM; for example, the newly developed immunotherapy agent, daratumumab, has been associated with a significant decrease in the number of collected HSC; in the CASSIOPEIA trial, 22% of the Dara-VTd group needed P during HSC mobilization, compared to 8% of the VTd group [51]. Finally, although the administration of P "just in time" allows a predictable increase of a factor ranging from x3-x4 the number of circulating CD34 +cells, the real increase after P cannot be easily predicted, leaving to the clinician the difficult final decision about whether or not to start the leukapheresis procedure after P administration.

References

[1] Hubel K, Fresen MM, Apperley JF, Basak GW, Douglas KW, Gabriel IH, et al. European data on stem cell mobilization with plerixafor in non–Hodgkin's lymphoma, Hodgkin's lymphoma and multiple myeloma patients. A subgroup analysis of the European Consortium of stem cell mobilization. Bone Marrow Transpl 2012;47:1046–50.

[2] Lin TL, Wang PN, Kuo MC, Hung YH, Chang H, Tang TC. Cyclophosphamide plus granulocyte-colony stimulating factor for hematopoietic stem cell mobilization in patients with multiple myeloma. J Clin Apher 2016;31:423–8.

[3] Anderlini P. Effects and safety of granulocyte colony-stimulating factor in healthy volunteers. Curr Opin Hematol 2009 Jan;16(1):35–44.

[4] LeBlanc R, Roy J, Vu L, Cantin G. A prospective study of G-CSF effects on hemostasis in allogeneic blood stem cell donors. Bone Marrow Transpl 1999;23: 991–6.

[5] Halter J, Kodera Y, Ispizua AU, Greinix HT, Schmitz N, Favre G, et al. Severe events in donors after allogeneic hematopoietic stem cell donation. Haematologica 2009; 94:94–101.

[6] Watts MJ, Addison I, Long SG, Hartley S, Warrington S, Boyce M, et al. Crossover study of the haematological effects and pharmacokinetics of glycosylated and non-glycosylated G-CSF in healthy volunteers. Br J Haematol 1997;98:474–9.

[7] Falanga A, Marchetti M, Evangelista V, Manarini S, Oldani E, Giovanelli S, et al. Neutrophil activation and hemostatic changes in healthy donors receiving granulocyte colony-stimulating factor. Blood 1999;93(8):2506–14.

[8] Canales MA, Arrieta R, Gomez-Rioja R, Diez J, Jimenez-Yuste V, Hernandez-Navarro F. Induction of a hypercoagulability state and endothelial cell activation by granulocyte colony-stimulating factor in peripheral blood stem cell donors. J Hematother Stem Cell Res 2002;11(4):675–81.

[9] Levesque J, Hendy J, Takamatsu Y, Simmons P, Bendall L. Disruption of the CXCR4/CXCL12 chemotactic interaction during hematopoietic stem cell mobilization induced by GCSF or cyclophosphamide. J Clin Invest 2003;111: 187–96.

[10] Saraceni F, Shem-Tov N, Olivieri A, Nagler A. Mobilized peripheral blood grafts include more than hematopoietic stem cells: the immunological perspective. Bone Marrow Transpl 2015;50(7):886–91.

[11] Olivieri A, Marchetti M, Lemoli R, Tarella C, Iacone A, Lanza F, et al. Proposed definition of 'poor mobilizer' in lymphoma and multiple myeloma: an analytic hierarchy process by ad hoc working group Gruppo ItalianoTrapianto di Midollo Osseo. Bone Marrow Transpl 2012;47(3):342–51.

[12] Wuchter P, Ran D, Bruckner T, Schmitt T, Witzens-Harig M, Neben K, et al. Poor mobilization of hematopoietic stem cells definitions, incidence, risk factors, and impact on outcome of autologous transplantation. Biol Blood Marrow Transpl 2010;16:490–9.

[13] Perseghin P, Terruzzi E, Dassi M, Baldini V, Parma M, Coluccia P, et al. Management of poor peripheral blood stem cell mobilization: incidence, predictive factors, alternative strategies and outcome. A retrospective analysis on 2177 patients from three major Italian institutions. Transfus Apher Sci 2009;41:33–7.

[14] Koenigsmann M, Jentsch-Ullrich K, Mohren M, Becker E, Heim M, Franke A. The role of diagnosis in patients failing peripheral blood progenitor cell mobilization. Transfusion 2004;44:777–84.

[15] Olivieri J, Attolico I, Nuccorini R, Pascale SP, Chiarucci M, Poiani M, et al. Predicting failure of hematopoietic stem cell mobilization before it starts: the predicted poor mobilizer (pPM) score. Bone Marrow Transpl 2018;53(4):461–73.

[16] Cao X, Wu X, Frassica D, Yu B, Pang L, Xian L, et al. Irradiation induces bone injury by damaging bone marrow microenvironment for stem cells. Proc Natl Acad Sci USA 2011;108:1609–14.

[17] Hooper AT, Butler JM, Nolan DJ, Kranz A, Iida K, Kobayashi M, et al. Engraftment and reconstitution of hematopoiesis is dependent on VEGFR2-mediated regeneration of sinusoidal endothelial cells. Cell Stem Cell 2009;4:263–74.

[18] Zhou BO, Yu H, Yue R, Zhao Z, Rios JJ, Naveiras O, et al. Bone marrow adipocytes promote the regeneration of stem cells and haematopoiesis by secreting SCF. Nat Cell Biol 2017;19:891–905.

[19] Fonseca-Pereira D, Arroz-Madeira S, Rodrigues-Campos M, Barbosa IAM, Domingues RG, Bento T, et al. The neurotrophic factor receptor RET drives haematopoietic stem cell survival and function. Nature 2014;514:98–101.

[20] Bensinger W, Appelbaum F, Rowley S, Storb R, Sanders J, Lilleby K, et al. Factors that influence collection and engraftment of autologous peripheral-blood stem cells. J Clin Oncol 1995;13:2547–55.

[21] Haas R, Mohle R, Fruhauf S, Goldschmidt H, Witt B, Flentje M, et al. Patient characteristics associated with successful mobilizing and autografting of peripheral blood progenitor cells in malignant lymphoma. Blood 1994;83:3787–94.

[22] Lucas D, Scheiermann C, Chow A, Kunisaki Y, Bruns I, Barrick C, et al. Chemotherapy-induced bone marrow nerve injury impairs hematopoietic regeneration. Nat Med 2013;19:695–703.

[23] Chen Jian, Lazarus Hillard M, Dahi Parastoo B, Avecilla Scott, Giralt Sergio A. Getting blood out of a stone: identification and management of patients with poor hematopoietic cell mobilization. Blood Rev 2021;47:10077.

[24] De la Rubia J, Blade J, Lahuerta JJ, Ribera JM, Martinez R, Alegre A, et al. Effect of chemotherapy with alkylating agents on the yield of CD34+ cells in patients with multiple myeloma. Results of the Spanish Myeloma Group (GEM)Study. Haematologica 2006;91:621–7.

[25] Boccadoro M, Palumbo A, Bringhen S, Merletti F, Ciccone G, Richiardi L, et al. Oral melphalan at diagnosis hampers adequate collection of peripheral blood progenitor cells in multiple myeloma. Haematologica 2002;87:846–50.

[26] Kemp K, Morse R, Wexler S, Cox C, Mallam E, Hows J, et al. Chemotherapy-induced mesenchymal stem cell damage in patients with hematological malignancy. Ann Hematol 2010;89:701–13.

[27] Paripati H, Stewart AK, Cabou S, Dueck A, Zepeda VJ, Pirooz N, et al. Compromised stem cell mobilization following induction therapy with lenalidomide in myeloma. Leukemia 2008;22:1282–4.

[28] Lysak D, Koza V, Steinerova K, Jindra P, Vozobulova V, Schutzova M. Mobilization of peripheral blood stem cells in CLL patients after front-line fludarabine treatment. Ann Hematol 2005;84:456–61.

[29] Moreau P, Attal M, Hulin C, Arnulf B, Belhadj K, Benboubker L, et al. Bortezomib, thalidomide, and dexamethasone with or without daratumumab before and after autologous stem-cell transplantation for newly diagnosed multiple myeloma (CASSIOPEIA): a randomised, open-label, phase 3 study. Lancet 2019;394:29–38.

[30] Morris CL, Siegel E, Barlogie B, Cottler-Fox M, Lin P, Fassas A, et al. Mobilization of CD34+ cells in elderly patients with multiple myeloma: influence of age, prior therapy, platelet count and mobilization regimen. Br J Haematol 2003;120: 413–23.
[31] Hosing C, Saliba RM, Ahlawat S, Korbling M, Kebriaei P, Alousi A, et al. Poor hematopoietic stem cell mobilizers: a single institution study of incidence and risk factors in patients with recurrent or relapsed lymphoma. Am J Hematol 2009;84: 335–7.
[32] Sauer S, Kriegsmann K, Nientiedt C, Schmitt A, Müller-Tidow C, Raab MS, Kauer J. Autologous stem cell collection after daratumumab, bortezomib, thalidomide, and dexamethasone versus bortezomib, cyclophosphamide, and dexamethasone in newly diagnosed multiple myeloma. Transfus Med Hemother 2023;50(5):371–81.
[33] Lenk J, Bornhauser M, Kramer M, Hölig K, Poppe-Thiede K, Schmidt H, et al. Sex and body mass index but Not CXCL12 801 G/A polymorphism determine the efficacy of hematopoietic cell mobilization: a study in healthy volunteer donors. Biol Blood Marrow Transplant 2013;19:1517–21.
[34] Szmigielska-Kaplon A, Szemraj J, Hamara K, Robak M, Wolska A, Pluta A, et al. Polymorphism of CD44 Influences the Efficacy of CD34+ cells mobilization in patients with hematological malignancies. Biol Blood Marrow Transplant 2014;20: 986–91.
[35] Shin S, Kim J, Kim-Wanner S-Z, Bönig H, Cho SR, Kim S, et al. A novel association between relaxin receptor polymorphism and hematopoietic stem cell yield after mobilization. PLOS ONE 2017;12:e0179986.
[36] Adamiak M, Poniewierska-Baran A, Borkowska S, Schneider G, Abdelbaset-Ismail A, Suszynska M, et al. Evidence that a lipolytic enzyme—hematopoietic-specific phospholipase C-β2—promotes mobilization of hematopoietic stem cells by decreasing their lipid raft-mediated bone marrow retention and increasing the promobilizing effects of granulocytes. Leukemia 2015;30:919–28.
[37] Fadini GP, Di Persio JF. Diabetes mellitus as a poor mobilizer condition. Blood Rev 2018;32:184–91.
[38] Gifford G, Hesson L, Wong JWH, Carroll A, Gabrielli S, Bai L, et al. Poor mobilization of autologous CD34+ peripheral blood stem cells in haematology patients undergoing autologous stem cell transplantation is associated with the presence of variants in genes implicated in clonal haematopoiesis of indeterminant potential. Br J Haematol 2021 May;193(4):841–4.
[39] Hazenberg CLE, de Graaf AO, Mulder R, Bungener LB, van Bergen MGJM, Mulder AB, et al. Clonal hematopoiesis in patients with stem cell mobilization failure: a nested case-control study. Blood Adv 2023;7(7):1269–78.
[40] Olivieri A, Marchetti M, Lemoli R, Tarella C, Iacone A, Lanza F, et al. Proposed definition of ‘poor mobilizer’ in lymphoma and multiple myeloma: an analytic hierarchy process by ad hoc working group Gruppo Italiano Trapianto di Midollo Osseo. Bone Marrow Transpl 2012;47:342–51.
[41] Pusic I, Jiang SY, Landua S, Uy GL, Rettig MP, Cashen AF, et al. Impact of mobilization and remobilization strategies on achieving sufficient stem cell yields for autologous transplantation. Biol Blood Marrow Transpl 2008;14:1045–56.
[42] Attolico I, Pavone V, Ostuni A, Rossini B, Musso M, Crescimanno A, et al. Plerixafor added to chemotherapy plus G–CSF is safe and allows adequate PBSC collection in predicted poor mobilizer patients with multiple myeloma or lymphoma. Biol Blood Marrow Transpl 2012;18:241–9.
[43] Farina L, Spina F, Guidetti A, Longoni P, Ravagnani F, Dodero A, et al. Peripheral blood CD34+ cell monitoring after cyclophosphamide and granulocyte-colony-stimulating factor: an algorithm for the pre-emptive use of plerixafor. Leuk Lymphoma 2014;55(2):331–6.
[44] D’Addio A, Curti A, Worel N, Douglas K, Motta MR, Rizzi S. The addition of plerixafor is safe and allows adequate PBSC collection in multiple myeloma and lymphoma patients poor mobilizers after chemotherapy and G-CSF. Bone Marrow Transpl 2011;46(3):356–63.
[45] Milone G, Tripepi G, Martino M, Ancora F, Bartolozzi B, Spadaro A, et al. Early measurement of CD34+ cells in peripheral blood after cyclophosphamide and granulocyte colony-stimulating factor treatment predicts later CD34+ mobilisation failure and is a possible criterion for guiding "on demand" use of plerixafor. Blood Transfus 2013;11(1):94–101.
[46] Sorasio R, Bonferroni M, Grasso M, Strola G, Rapezzi D, Marenchino D, et al. Peripheral blood CD34+ percentage at hematological recovery after chemotherapy is a good early predictor of harvest: a single-center experience. Biol Blood Marrow Transpl 2014;20(5):717–23.
[47] Chow E, Rao KV, Wood WA, Covington D, Armistead PM, Coghill J, et al. Effectiveness of an algorithm-based approach to the utilization of plerixafor in patients undergoing chemotherapybased stem cell mobilization. Biol Blood Marrow Transpl 2014;20:1064–8.
[48] Costa LJ, Alexander ET, Hogan KR, Schaub C, Fouts TV, Stuart RK. Development and validation of a decision-making algorithm to guide the use of plerixafor for autologous hematopoietic stem cell mobilization. Bone Marrow Transpl 2011;6: 64–9.
[49] Mohty M, Hübel K, Kröger N, Aljurf M, Apperley J, Basak GW, et al. Autologous haematopoietic stem cell mobilisation in multiple myeloma and lymphoma patients: a position statement from the European group for blood and marrow transplantation. Bone Marrow Transpl 2014;49:865–72.
[50] Giralt S, Costa L, Schriber J, Dipersio J, Maziarz R, McCarty J, et al. Optimizing autologous stem cell mobilization strategies to improve patient outcomes: consensus guidelines and recommendations. Biol Blood Marrow Transpl 2014;20: 295–308.
[51] Mina R, Petrucci MT, Bonello F, Bongarzoni V, Saccardi R, Bertuglia G, et al. A prospective, multicenter study on hematopoietic stem-cell mobilization with cyclophosphamide plus granulocyte colony-stimulating factor and ’on-demand’ plerixafor in multiple myeloma patients treated with novel agents. Haematologica 2023. https://doi.org/10.3324/haematol.2023.284023.